

postgres/pg_create_subscription@PostgreSQL 18* X

postgres/postgres@PostgreSQL 16



Query Query History

```
1 CREATE TABLE airline_info (  
2     airline_id INTEGER PRIMARY KEY NOT NULL,  
3     airline_code VARCHAR(30) NOT NULL,  
4     airline_name VARCHAR(50) NOT NULL,  
5     airline_country VARCHAR(50) NOT NULL,  
6     created_at TIMESTAMP NOT NULL,  
7     updated_at TIMESTAMP NOT NULL,  
8     info VARCHAR(50) NOT NULL  
9 );
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 285 msec.

postgres/pg_create_subscription@PostgreSQL 18* X

postgres/postgres@PostgreSQL 16

A toolbar containing various icons for file operations (like save, open, delete), editing (like undo, redo, find), and execution (like run, stop, refresh). It also includes a dropdown menu for 'No limit' and a help icon.

Query Query History

```
1 CREATE TABLE airport(  
2     airport_id INTEGER PRIMARY KEY NOT NULL,  
3     airport_name VARCHAR(50) NOT NULL,  
4     country VARCHAR(50) NOT NULL,  
5     state VARCHAR(50) NOT NULL,  
6     city VARCHAR(50) NOT NULL,  
7     created_at TIMESTAMP NOT NULL,  
8     updated_at TIMESTAMP NOT NULL  
9 );
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 87 msec.

Total rows: Query complete 20:22:22.227

pgAdmin 4

File Object Tools Edit View Window Help



postgres/pg_create_subscription@PostgreSQL 18* X



postgres/postgres@PostgreSQL 16



No limit



Query Query History

```
1
2 CREATE TABLE baggage_check(
3     baggage_check_id INTEGER PRIMARY KEY NOT NULL,
4     check_result VARCHAR(50) NOT NULL,
5     created_at TIMESTAMP NOT NULL,
6     updated_at TIMESTAMP NOT NULL,
7     booking_id INTEGER NOT NULL,
8     passenger_id INTEGER NOT NULL
9 )
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 86 msec.



postgres/pg_create_subscription@PostgreSQL 18* X



postgres/postgres@PostgreSQL 16



No limit



Query

Query History



1

2

3

4

5

6

7

8

```
CREATE TABLE baggage(  
    baggage_id INTEGER PRIMARY KEY NOT NULL,  
    weight_in_kg DECIMAL(4,2) NOT NULL,  
    created_at TIMESTAMP NOT NULL,  
    updated_at TIMESTAMP NOT NULL,  
    booking_id INTEGER  
);
```

Data Output

Messages

Notifications

CREATE TABLE

Query returned successfully in 92 msec.

postgres/pg_create_subscription@PostgreSQL 18* X

postgres/postgres@PostgreSQL 16

No limit

Query Query History

```
1  
2 CREATE TABLE boarding_pass(  
3     boarding_pass_id INTEGER PRIMARY KEY NOT NULL,  
4     booking_id INTEGER NOT NULL,  
5     seat VARCHAR(50) NOT NULL,  
6     boarding_time TIMESTAMP NOT NULL,  
7     created_at TIMESTAMP NOT NULL,  
8     updated_at TIMESTAMP NOT NULL  
9 );
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 84 msec.

postgres/pg_create_subscription@PostgreSQL 18* X

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Query Query History

```
1  
2 CREATE TABLE booking_flight(  
3     booking_flight_id INTEGER PRIMARY KEY NOT NULL,  
4     booking_id INTEGER NOT NULL,  
5     flight_id INT NOT NULL,  
6     created_at TIMESTAMP NOT NULL,  
7     updated_at TIMESTAMP NOT NULL  
8 );
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 81 msec.



postgres/pg_create_subscription@PostgreSQL 18* X



postgres/postgres@PostgreSQL 16



No limit



Query Query History

1
2
3
4
5
6
7
8
9
10
11

```
CREATE TABLE booking(  
    booking_id INT PRIMARY KEY NOT NULL,  
    flight_id INT NOT NULL,  
    passenger_id INT NOT NULL,  
    booking_platform VARCHAR(50) NOT NULL,  
    created_at TIMESTAMP NOT NULL,  
    updated_at TIMESTAMP NOT NULL,  
    status VARCHAR(50) NOT NULL,  
    price DECIMAL(7,2) NOT NULL  
);
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 78 msec.

postgres/pg_create_subscription@PostgreSQL 18* X

postgres/postgres@PostgreSQL 16

No limit

Execute script

F5

Query Query History

```
1
2 CREATE TABLE flights(
3     flight_id INT PRIMARY KEY NOT NULL,
4     sch_departure_time TIMESTAMP NOT NULL,
5     sch_arrival_time TIMESTAMP NOT NULL,
6     departing_airport_id INT NOT NULL,
7     arriving_airport_id INT NOT NULL,
8     departing_gate VARCHAR(50) NOT NULL,
9     arriving_gate VARCHAR(50) NOT NULL,
10    airline_id INT NOT NULL,
11    act_departure_time TIMESTAMP NOT NULL,
12    act_arrival_time TIMESTAMP NOT NULL,
13    created_at TIMESTAMP NOT NULL,
14    updated_at TIMESTAMP NOT NULL
15 );
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 89 msec.



postgres/pg_create_subscription@PostgreSQL 18* X



postgres/postgres@PostgreSQL 16



No limit



Query

Query History

1
2
3
4
5
6
7
8
9
10
11
12
13

```
CREATE TABLE passengers(  
    passengers_id INT PRIMARY KEY NOT NULL,  
    first_name VARCHAR(50) NOT NULL,  
    last_name VARCHAR(50) NOT NULL,  
    date_of_birth DATE NOT NULL,  
    gender VARCHAR(50) NOT NULL,  
    country_of_citizenship VARCHAR(50) NOT NULL,  
    country_of_residence VARCHAR(50) NOT NULL,  
    passport_number VARCHAR(20) NOT NULL,  
    created_at TIMESTAMP NOT NULL,  
    updated_at TIMESTAMP NOT NULL  
);
```

Execute script

F5

Data Output

Messages

Notifications

CREATE TABLE

postgres/pg_create_subscription@PostgreSQL 18* X



postgres/postgres@PostgreSQL 16



Query Query History

```
1
2 CREATE TABLE security_check(
3     security_check_id INT PRIMARY KEY NOT NULL,
4     check_result VARCHAR(20) NOT NULL,
5     created_at TIMESTAMP NOT NULL,
6     updated_at TIMESTAMP NOT NULL,
7     passenger_id INT NOT NULL
8 );
```

Data Output Messages Notifications

CREATE TABLE



postgres/pg_create_subscription@PostgreSQL 18* X



postgres/postgres@PostgreSQL 16



No limit



Query

Query History



1

ALTER TABLE airline_info

2

RENAME TO airline;

Execute script

F5

Data Output

Messages

Notifications

ALTER TABLE

Query returned successfully in 94 msec.

postgres/pg_create_subscription@PostgreSQL 18* X

postgres/postgres@PostgreSQL 16

A row of icons for pgAdmin 4, including a folder, a save icon, a pencil, a funnel, a dropdown menu with 'No limit', a play button, a refresh button, a search icon, a table icon, a dropdown menu, a refresh icon, a refresh icon, a list icon, and a help icon.

Query Query History

```
1 ALTER TABLE flights
2 ALTER COLUMN departing_gate TYPE TEXT;
```

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 65 msec.

postgres/pg_create_subscription@PostgreSQL 18* X

postgres/postgres@PostgreSQL 16



Query Query History

```
1 ALTER TABLE airline
2 DROP COLUMN info;
```

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 77 msec.

postgres/pg_create_subscription@PostgreSQL 18* X

postgres/postgres@PostgreSQL 16



Query Query History

```
1 ALTER TABLE booking
2 ADD CONSTRAINT booking_passenger_fk
3 FOREIGN KEY (passenger_id)
4 REFERENCES passengers(passenger_id);
5 ALTER TABLE security_check
6 ADD CONSTRAINT security_check_passenger_fk
7 FOREIGN KEY (passenger_id)
8 REFERENCES passengers(passenger_id);
9 ALTER TABLE baggage_check
10 ADD CONSTRAINT baggage_check_passenger_fk
11 FOREIGN KEY (passenger_id)
12 REFERENCES passengers(passenger_id);
```

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 69 msec.

postgres/pg_create_subscription@PostgreSQL 18* X

postgres/postgres@PostgreSQL 16

A toolbar containing icons for file operations (new, open, save, print, copy, paste) and query execution (execute, stop, refresh, etc.).

Query Query History

```
1 ALTER TABLE baggage_check
2 ADD CONSTRAINT baggage_check_booking_fk
3 FOREIGN KEY (booking_id)
4 REFERENCES booking(booking_id);
5 ALTER TABLE baggage
6 ADD CONSTRAINT baggage_booking_fk
7 FOREIGN KEY (booking_id)
8 REFERENCES booking(booking_id);
9 ALTER TABLE boarding_pass
10 ADD CONSTRAINT boarding_pass_booking_fk
11 FOREIGN KEY (booking_id)
12 REFERENCES booking(booking_id);
13 ALTER TABLE booking_flight
14 ADD CONSTRAINT booking_flight_booking_fk
15 FOREIGN KEY (booking_id)
16 REFERENCES booking(booking_id);
```

Data Output Messages Notifications

ERROR: ограничение "baggage_check_booking_fk" для отношения "baggage_check" уже существует

ОШИБКА: ограничение "baggage_check_booking_fk" для отношения "baggage_check" уже существует
SQL state: 42710

postgres/pg_create_subscription@PostgreSQL 18* X

postgres/postgres@PostgreSQL 16



Query Query History

```
1 ALTER TABLE booking_flight
2 ADD CONSTRAINT booking_flight_flights_fk
3 FOREIGN KEY (flight_id)
4 REFERENCES flights(flight_id);
```

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 120 msec.

postgres/pg_create_subscription@PostgreSQL 18* X

postgres/postgres@PostgreSQL 16



No limit



Query

Query History

```
1 ALTER TABLE flights
2 ADD CONSTRAINT flights_departing_airport_fk
3 FOREIGN KEY (departing_airport_id)
4 REFERENCES airport(airport_id);
```

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 71 msec.

postgres/pg_create_subscription@PostgreSQL 18* X

postgres/postgres@PostgreSQL 16

A row of icons for file operations (folder, save, edit), filtering (funnel), and execution (play, stop, refresh, etc.).

Query Query History

```
1 ALTER TABLE flights
2 ADD CONSTRAINT flights_arriving_airport_fk
3 FOREIGN KEY (arriving_airport_id)
4 REFERENCES airport(airport_id);
```

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 83 msec.



postgres/pg_create_subscription@PostgreSQL 18* X



postgres/postgres@PostgreSQL 16



No limit



Query Query History

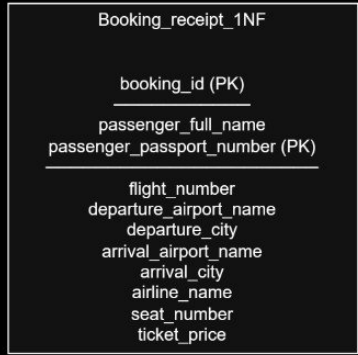


```
1 ALTER TABLE flights
2 ADD CONSTRAINT flights_airline_fk
3 FOREIGN KEY (airline_id)
4 REFERENCES airport(airport_id);
```

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 64 msec.

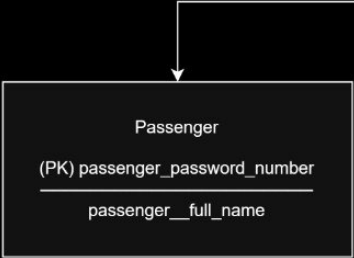


To achieve First Normal Form (1NF), the seat_numbers attribute is changed to seat_number so that each cell contains only one atomic value. Moreover this table has not FK.

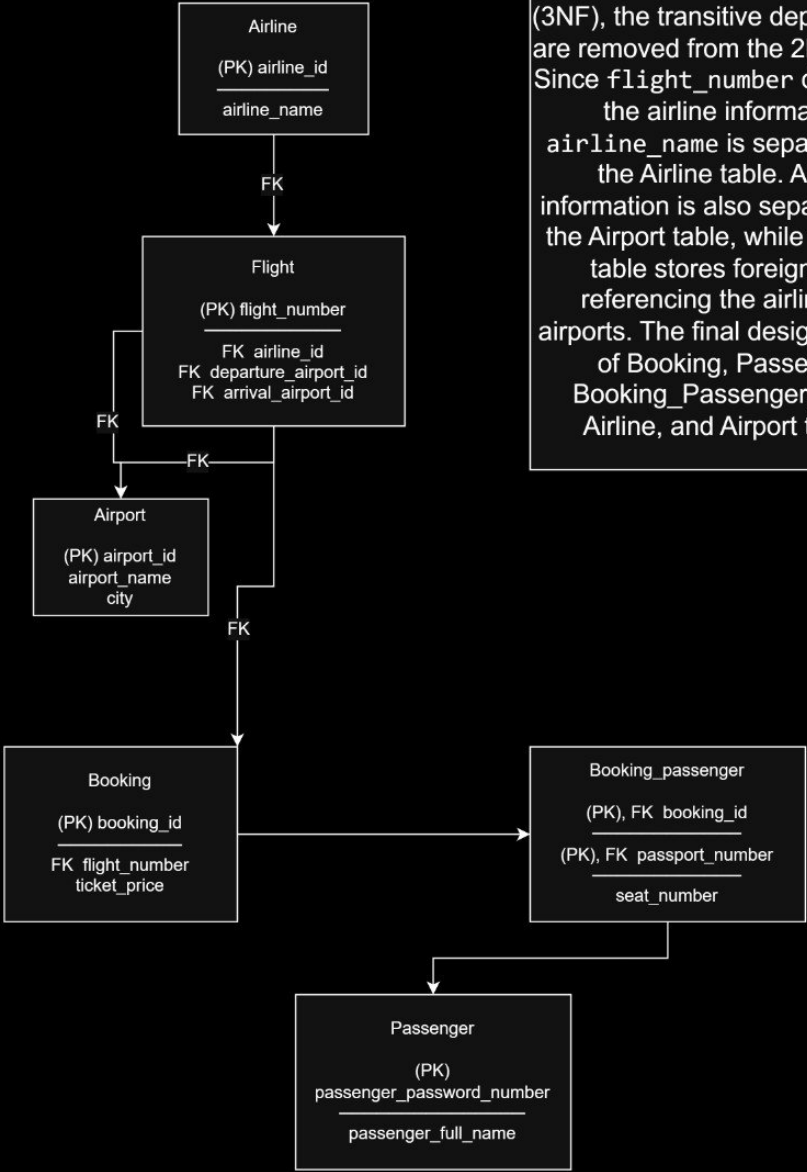
2NF

3NF

To achieve Third Normal Form (3NF), the transitive dependencies are removed from the 2NF design. Since flight_number determines the airline information, airline_name is separated into the Airline table. Airport information is also separated into the Airport table, while the Flight table stores foreign keys referencing the airline and airports. The final design consists of Booking, Passenger, Booking_Passenger, Flight, Airline, and Airport tables.



To achieve Second Normal Form (2NF), the partial dependencies from the 1NF table are removed. flight_number and ticket_price depend on booking_id, while passenger_full_name depends on passenger_passport_number. Therefore, the data is separated into Booking, Passenger, and Booking_Passenger tables. The Booking_Passenger table uses a composite primary key consisting of booking_id and passenger_passport_number, which are also foreign keys.



postgres/pg_create_subscription@PostgreSQL 18+ X

postgres/postgres@PostgreSQL 16

    No limit      

Query Query History

```
1 CREATE TABLE norm_airline (  
2     airline_id INTEGER PRIMARY KEY,  
3     airline_name VARCHAR(50) NOT NULL  
4 );  
5 CREATE TABLE norm_airport (  
6     airport_id INTEGER PRIMARY KEY,  
7     airport_name VARCHAR(50) NOT NULL,  
8     city VARCHAR(50) NOT NULL  
9 );  
10 CREATE TABLE norm_flight (  
11     flight_number VARCHAR(20) PRIMARY KEY,  
12     airline_id INTEGER NOT NULL,  
13     departure_airport_id INTEGER NOT NULL,  
14     arrival_airport_id INTEGER NOT NULL,  
15     FOREIGN KEY (airline_id)  
16         REFERENCES norm_airline(airline_id),
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 94 msec.



postgres/pg_create_subscription@PostgreSQL 18* X



postgres/postgres@PostgreSQL 16



Query Query History



```
16 REFERENCES norm_airline(airline_id),
17 FOREIGN KEY (departure_airport_id)
18 REFERENCES norm_airport(airport_id),
19 FOREIGN KEY (arrival_airport_id)
20 REFERENCES norm_airport(airport_id)
21 );
22 CREATE TABLE norm_booking (
23     booking_id INTEGER PRIMARY KEY,
24     flight_number VARCHAR(20) NOT NULL,
25     ticket_price DECIMAL(7,2) NOT NULL,
26     FOREIGN KEY (flight_number)
27     REFERENCES norm_flight(flight_number)
28 );
29 CREATE TABLE norm_passenger (
30     passenger_passport_number VARCHAR(20) PRIMARY KEY,
31     passenger_full_name VARCHAR(100) NOT NULL
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 94 msec.

postgres/pg_create_subscription@PostgreSQL 18* X

postgres/postgres@PostgreSQL 16

A row of icons for various database operations: folder, save, edit, filter, a dropdown menu showing 'No limit', and several other icons for query execution and management.

Query Query History

```
27         REFERENCES norm_flight(flight_number)
28     );
29     CREATE TABLE norm_passenger (
30         passenger_passport_number VARCHAR(20) PRIMARY KEY,
31         passenger_full_name VARCHAR(100) NOT NULL
32     );
33     CREATE TABLE norm_booking_passenger (
34         booking_id INTEGER NOT NULL,
35         passenger_passport_number VARCHAR(20) NOT NULL,
36         seat_number VARCHAR(10) NOT NULL,
37         PRIMARY KEY (booking_id, passenger_passport_number),
38         FOREIGN KEY (booking_id)
39             REFERENCES norm_booking(booking_id),
40         FOREIGN KEY (passenger_passport_number)
41             REFERENCES norm_passenger(passenger_passport_number)
42     );
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 94 msec.



postgres/pg_create_subscription@PostgreSQL 18* X



postgres/postgres@PostgreSQL 16



No limit



Query Query History



```
1  INSERT INTO norm_airport
2  (airport_id, airport_name, city)
3  VALUES
4  (1, 'Almaty International Airport', 'Almaty'),
5  (2, 'Istanbul Airport', 'Istanbul'),
6  (3, 'Astana International Airport', 'Astana'),
7  (4, 'Frankfurt Airport', 'Frankfurt'),
8  (5, 'Hamad International Airport', 'Doha');
```

Data Output Messages Notifications

INSERT 0 5

Query returned successfully in 88 msec.



postgres/pg_create_subscription@PostgreSQL 18* X



postgres/postgres@PostgreSQL 16



No limit



Query Query History



```
1  INSERT INTO norm_flight
2  (flight_number, airline_id, departure_airport_id, arrival_airport_id)
3  VALUES
4  ('KC101', 1, 1, 2),
5  ('TK350', 2, 2, 1),
6  ('KC201', 3, 3, 1),
7  ('LH600', 4, 1, 4),
8  ('QR390', 5, 1, 5);
```

Data Output Messages Notifications

INSERT 0 5

Query returned successfully in 112 msec.

postgres/pg_create_subscription@PostgreSQL 18* X

postgres/postgres@PostgreSQL 16



Query Query History

```
1  INSERT INTO norm_booking
2  (booking_id, flight_number, ticket_price)
3  VALUES
4  (1, 'KC101', 500.00),
5  (2, 'TK350', 450.00),
6  (3, 'KC201', 300.00),
7  (4, 'LH600', 700.00),
8  (5, 'QR390', 650.00);
```

Data Output Messages Notifications

INSERT 0 5

Query returned successfully in 71 msec.



postgres/pg_create_subscription@PostgreSQL 18* X



postgres/postgres@PostgreSQL 16



No limit



Query

Query History



```
1  INSERT INTO norm_passenger
2  (passenger_passport_number, passenger_full_name)
3  VALUES
4  ('P001', 'John Smith'),
5  ('P002', 'Anna Brown'),
6  ('P003', 'Ali Karimov'),
7  ('P004', 'Maria Ivanova'),
8  ('P005', 'David Lee');
```

Data Output

Messages

Notifications

INSERT 0 5

Query returned successfully in 76 msec.



postgres/pg_create_subscription@PostgreSQL 18* X



postgres/postgres@PostgreSQL 16



No limit



Query

Query History

```
1  INSERT INTO norm_booking_passenger
2  (booking_id, passenger_passport_number, seat_number)
3  VALUES
4  (1, 'P001', '12A'),
5  (2, 'P002', '14B'),
6  (3, 'P003', '8C'),
7  (4, 'P004', '20A'),
8  (5, 'P005', '15D');
```

Data Output

Messages

Notifications

INSERT 6 5

Query returned successfully in 66 msec.



postgres/pg_create_subscription@PostgreSQL 18* X



postgres/postgres@PostgreSQL 16



No limit



Query Query History



```
1 SELECT * FROM norm_airline;
2 SELECT * FROM norm_airport;
3 SELECT * FROM norm_flight;
4 SELECT * FROM norm_booking;
5 SELECT * FROM norm_passenger;
6 SELECT * FROM norm_booking_passenger;
```

Data Output Messages Notifications



SQL

Show

	booking_id [PK] integer	passenger_passport_number [PK] character varying (20)	seat_number character varying (10)
1	1	P001	12A
2	2	P002	14B
3	3	P003	8C
4	4	P004	20A
5	5	P005	15D